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Expt.No: 6

Title: Numerical Processing using Numpy

```
In [1]: 1 import numpy as np
2
3 array = np.array([1, 2, 3, 4, 5])
4 print("Original Array:")
5 print(array)
6
7 array_add = array + 5
8 array_sub = array - 2
9 array_mul = array * 3
10 array_div = array / 2
11 array_sqr = array ** 2
12
13 print("\nArithmetic Operations:")
14 print("Add 5: ", array_add)
15 print("Subtract 2:", array_sub)
16 print("Multiply by 3:", array_mul)
17 print("Divide by 2:", array_div)
18 print("Square:", array_sqr)
19
20 array_sqrt = np.sqrt(array)
21 array_exp = np.exp(array)
22
23 print("\nUniversal Functions:")
24 print("Square Root:", array_sqrt)
25 print("Exponential:", array_exp)
26
27 mean = np.mean(array)
28 std_dev = np.std(array)
29 sum_array = np.sum(array)
30
31 print("\nStatistical Operations:")
32 print("Mean:", mean)
33 print("Standard Deviation:", std_dev)
34 print("Sum:", sum_array)
```

Original Array:

[1 2 3 4 5]

Arithmetic Operations:

Add 5: [6 7 8 9 10]

Subtract 2: [-1 0 1 2 3]

Multiply by 3: [3 6 9 12 15]

Divide by 2: [0.5 1. 1.5 2. 2.5]

Square: [1 4 9 16 25]

Universal Functions:

Square Root: [1. 1.41421356 1.73205081 2. 2.23606798]

Exponential: [2.71828183 7.3890561 20.08553692 54.59815003 148.4131591]

Statistical Operations:

Mean: 3.0

Standard Deviation: 1.4142135623730951

Sum: 15

In []:

1

